

Introduction to Regression in Machine Learning

1. What is Regression in Machine Learning?

Regression is a supervised machine learning method that uses one or more input variables (features) to predict continuous numerical outcomes. Regression is to determine the link between independent variables (predictors) and a dependent variable (target) in order to precisely forecast future numerical outcomes.

Regression forecasts numbers like prices, temperatures, sales, or income, in contrast to other machine learning techniques that forecast categories. It is extensively utilized in business, engineering, finance, healthcare, and numerous other domains where numerical value predicting is crucial.

Difference between Regression and Classification

Although both regression and classification belong to supervised learning, they solve different types of problems.

- **Regression** predicts continuous numerical values.
- **Classification** predicts discrete categories or class labels.

For example, predicting the selling price of a used car is a regression problem because the output is a number (e.g., \$12,500). Predicting whether an email is spam or not spam is a classification problem because the output belongs to one of two categories.

Examples

Regression

Example:

A real estate company predicts house prices based on house size, number of bedrooms, location, and age of the house.

Classification

Example:

A bank predicts whether a loan applicant will default on a loan (Yes or No).

2. Types of Regression

2.1 Linear Regression

Linear Regression is the simplest regression algorithm. It models a straight-line relationship between one independent variable and one dependent variable.

Basic Idea

The model fits the best straight line through the data using the equation:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where:

- Y = predicted value
- X = independent variable
- β_0 = intercept
- β_1 = slope
- ε = error term

Real-World Use Case

Predicting employee salary based on years of work experience.

Advantages

- Easy to understand and interpret.
- Fast to train.
- Works well when relationships are linear.

Limitations

- Cannot model complex nonlinear relationships.
- Sensitive to outliers.
- Assumes linearity between variables.

2.2 Multiple Linear Regression

Multiple Linear Regression extends linear regression by using two or more independent variables to predict one continuous target variable.

Basic Idea

The equation becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Each predictor contributes to estimating the target variable.

Real-World Use Case

Predicting house prices using:

- House size
- Number of bedrooms
- Location
- Age of the house

Advantages

- More accurate than simple linear regression.
- Considers multiple influencing factors.
- Widely used in business analytics.

Limitations

- More complex.
- Can suffer from multicollinearity.
- Requires careful feature selection.

2.3 Polynomial Regression

Polynomial Regression is used when the relationship between variables is nonlinear. It transforms input variables into polynomial features while still using linear regression techniques.

Basic Idea

Instead of fitting a straight line, the model fits a curve using polynomial terms.

Example:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$$

Real-World Use Case

Predicting crop yield based on rainfall where the relationship is curved rather than linear.

Advantages

- Captures nonlinear patterns.
- Often improves prediction accuracy.
- Flexible for complex datasets.

Limitations

- Easily overfits the training data.
- Harder to interpret.
- High-degree polynomials may reduce generalization.

3. Regression Metrics

Regression models are evaluated using several performance metrics.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between predicted and actual values.

Formula:

$$\text{MAE} = (1/n) \sum |\text{Actual} - \text{Predicted}|$$

A smaller MAE indicates better model performance.

Mean Squared Error (MSE)

MSE measures the average squared difference between predictions and actual values.

Formula:

$$\text{MSE} = (1/n) \sum (\text{Actual} - \text{Predicted})^2$$

Because errors are squared, large errors receive a much greater penalty.

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE is easier to interpret because it has the same units as the target variable.

Coefficient of Determination (R²)

R² measures how much variation in the target variable is explained by the model.

Formula:

$$R^2 = 1 - (\text{Residual Sum of Squares} / \text{Total Sum of Squares})$$

An R² value closer to 1 indicates a better model fit.

Comparison of Regression Metrics

Metric	Measures	Units	Sensitive to Large Errors	Interpretation
MAE	Average absolute error	Same as target	Low	Lower is better
MSE	Average squared error	Squared units	Very High	Lower is better
RMSE	Square root of MSE	Same as target	High	Lower is better
R^2	Explained variance	No units	No	Closer to 1 is better

4. Underfitting and Overfitting

Underfitting

Underfitting occurs when a regression model is too simple to capture the relationship between the input variables and the target variable. As a result, the model performs poorly on both training and testing data.

Example:

Using a straight line to model data that actually follows a curved pattern.

Overfitting

Overfitting occurs when a model learns not only the true patterns but also the noise in the training data. Such models achieve excellent training accuracy but perform poorly on unseen data.

Polynomial regression is especially vulnerable to overfitting when a very high polynomial degree is used.

Causes of Overfitting

- Using a polynomial degree that is too high.
- Small training datasets.
- Too many predictor variables.

- Presence of noise in the data.

Methods to Prevent Overfitting

1. Use cross-validation to evaluate model performance.
2. Apply regularization techniques such as Ridge or Lasso Regression.
3. Reduce model complexity by selecting an appropriate polynomial degree or removing unnecessary features.

5. Real-World Case Study

Predicting House Prices Using Multiple Linear Regression

One of the most common applications of regression is predicting residential property prices.

Goal

The objective is to estimate house prices accurately to assist buyers, sellers, and real estate agencies in making informed decisions.

Data Used

The dataset typically includes:

- House size (square meters)
- Number of bedrooms
- Number of bathrooms
- Age of the house
- Neighborhood or location
- Distance to schools and shopping centers

Regression Type

Researchers commonly use **Multiple Linear Regression** because house prices are influenced by multiple independent variables simultaneously.

Key Results

The study found that house size and location were the strongest predictors of price. The regression model explained a high percentage of the variation in housing prices (high R^2 value), allowing real estate companies to estimate market values more accurately. These predictions support better pricing strategies, investment decisions, and mortgage assessments.

In conclusion

One of the most crucial supervised machine learning methods for forecasting continuous numerical values is regression. Various regression techniques, such as Polynomial Regression, Multiple Linear Regression, and Linear Regression, are appropriate for various kinds of variable connections. Building trustworthy predicting systems requires choosing the right regression model and assessing it using metrics like MAE, MSE, RMSE, and R^2 . Understanding underfitting and overfitting also guarantees that predictions generalize well to fresh data and enhances model performance. Regression continues to play a significant role in solving real-world problems across business, healthcare, transportation, education, and many other domains.